

Night tubes and crime - progress outline

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1 Introduction

Over the second half of 2016, Transport for London phased in the Night Tube over a number of lines. This was an extension of a line's regular tube service through the night on Fridays and Saturdays, rather than ending at 12am and restarting the next morning at 5.30am. The purpose of the Night Tube was to allow Londoners to spend more time enjoying London's night-time economy, but as a consequence it means many people are walking around tube stations in the dark, often alone, and often without their full mental capacity. In fact, in the eyes of a criminal, the areas around tube stations become hotspots for these vulnerable targets, seemingly presenting a perfect opportunity to commit a crime. This is dangerous, and determining whether criminals acted upon this possibility will have significant implications on urban crime policy, as well as giving information on how criminals behave. I am therefore interested in determining the effect the opening of a night-tube line had on the prevalence and nature of crime in the areas surrounding the stations.

I am also interested in questions about the evolution and origin of the effect. The questions I want to answer are the following:

- Did the introduction of the night tubes have an effect on crime frequency and nature in areas surrounding the stations?
- How does this effect vary according to distance from the station? In particular, is there evidence to suggest criminals are responsive to nearby opportunities, or do they face spatial frictions?
- How do the effects evolve in time? Do criminals learn, and respond accordingly?
- If effects are present, are they coming from substitution away from crime elsewhere, or are they newly generated crimes?

To evaluate the overall effects of the policy, I am also interested in the house price response to the opening of the new stations, since in the short-term they proxy for regional 'attractiveness'. I might therefore also address the following question: to what extent are the benefits of late-night access cancelled out by the danger associated with travelling home afterwards? I don't know if I'll do this though.

2 Context

The Night Tube service, announced in November 2013 and first introduced in August 2016, is the first and only attempt Transport for London (TfL) have made at delivering overnight underground travel. Prior to this, tube services ended around midnight, and began around

5.30am the next morning; the Night Tube extended the service through the night on Fridays and Saturdays, in order to fill this gap. Its stated goal was to support London's night-time economy, by providing residents with an accessible option to get home and therefore encouraging them to enjoy the city's night-time amenities to their full extent.

The Night Tube service was gradually rolled out across different London Underground lines over the second half of 2016, as shown in Table 1.

Table 1: Night Tube introduction month and year by London Tube line

Line	Introduction Date
Bakerloo	Never
Central	August 2016
Circle	Never
District	Never
Hammersmith & City	Never
Jubilee	October 2016
Metropolitan	Never
Northern	November 2016
Piccadilly	December 2016
Victoria	August 2016
Waterloo & City	Never

Some of the most important contextual facts about the Night Tube rollout that might appear to pose an issue with identification, along with how I justify they aren't really big issues, are as follows:

- The lines that got the service were chosen on the basis of existing demand for overnight transport
 - This is an issue for my event studies if higher demand for a route leads to differential *evolution* of crimes after the Night tube introduction as compared to before its introduction. This doesn't seem like a huge issue
 - Also, many lines were excluded for other reasons anyway; often ongoing upgrades that required night-time work.
 - Either way, the pre-trends in my estimates look fine, so I don't think this is really an issue
- Trains on the lines arrived at slightly different frequencies - on the main (central) stretch of each line, they arrived at roughly 10 minute intervals, but there are segments of the Central and Northern lines, towards the ends, where trains arrived every 15 and 20 minutes respectively
 - I am slightly concerned this means treatment is not homogeneous across stations. However, I think that I can defend defining treatment uniformly across the Night Tube stations in two ways. Firstly, I can just think of treatment as on the extensive margin, i.e. 'being near a night tube station', and then the treatment effect just being potentially heterogeneous across areas around stations according to service frequency. Secondly, on the intensive margin I am more interested in the change in night-time ridership induced by the night tube, and I would presume many people in places with less frequent services would be happy to wait a little longer or time their journeys better in order to get the tube anyway - there might not be such a large drop-off in ridership.

- On the whole, I’m not really too worried about this
- TfL were aware of the implications the Night Tube might have on crime in surrounding areas, and therefore introduced additional policing measures to deal with this possibility. All Night Tube stations operated with full staffing, and police presence on the tubes themselves was substantially increased. Alongside the stations themselves, officers were also instructed to patrol a selection of the immediate key transport hubs and connecting bus stations. In particular, TfL identified 12 stations as “red stations”, predominantly in central London, where crime was expected to increase most, and these stations received enhanced police presence.
 - This is a bit of a concern from a reporting point of view - if there are more police, it is easier to report a crime, so there may be measurement error in my crime counts.
 - I think I can deal with this issue by showing that there are categories of serious crimes (i.e. ones that would have been reported anyway) that increase around the stations, which there are. Also, since the police presence was almost completely at the station, there should be no ease of reporting effect for crimes beyond a certain distance from the station, but we still see positive effects a long way out. Also, even if there is overreporting, an increase is indicative of the fact that there is an increase in crimes in the first place (otherwise there is nothing extra to report).
 - Also the presence of police may dampen the crime effect. Without this additional police presence, the effect might be larger. But this doesn’t really seem like a problem.
 - On the whole, I’ve convinced myself this isn’t as much of a problem as it seems
- In December 2017, the Night Tube was extended to certain parts of the London Overground. This is why I don’t extend my sample any further than 2017

It is worth noting that the stations served, and the service frequencies, did not change between the start of the service and the end of the sample period. Treatment is therefore consistent over time, which is good.

3 Literature

I have found quite a lot of useful literature relevant to this question, but the most similar paper, by a long way, is Zhang et al (2022). This paper also investigates the effect of the introduction of the Night Tube on a variety of social outcomes, including crime and house prices. They provide useful analysis that suggests crimes do increase as a result of the policy in surrounding areas, along with house prices, and demonstrates effects for other social outcomes too. However, their focus is on the estimator; they use double ML to estimate simple 2x2 difference in differences regressions. They don’t explain the crime effect - or any other effects - very deeply, and they don’t deal with many of the questions I am interested in answering, such as the dynamics of the effect in time and space, and the mechanisms behind the effect. They also use much less temporally and spatially granular data, which limits their ability to convincingly answer these questions anyway.

I’m wary of examining the same outcome in the same natural experiment as another paper, because I want to eventually publish this one. However, I think my contribution would be distinct from theirs in the following ways:

- Their focus is largely on their estimator, and they spend very little time discussing the crime effect, stopping with not much more than the overall static effect from their 2x2 DiD. My analysis focuses directly on crime and will therefore be a lot more detailed - I will address in depth questions that they didn't, including the mechanism and the spatial and temporal evolution of the effect, as well as whether the crime is new or not.
- I also seek to provide a contribution to the literature on the economic underpinnings of criminal behaviour, using this setting to comment on patterns of crime more generally. In this sense the paper would belong largely to the field of crime economics, rather than urban economics, which Zhang et al (2022) certainly lies within.
- I would also be extending their identification strategy to deal with staggered treatment and dynamic treatment effects in time and space, and will use a lot more granular data. This allows me to make further policy-relevant comments on the effect of the Night Tube, and in this sense provide much more detail to the conclusions they originally made. I therefore also see this as a novel contribution to urban economics, as well as the economics of crime.

However, my reservation about including the house price analysis is because of the fact that this paper also investigated the house price effect of the policy. I think if I did do that the paper would just become too similar to theirs to be thought of as a novel contribution that could eventually be published in a good journal.

I think this paper contributes to both crime and urban economics. To urban economics, the paper studies in detail one of the major unintended consequences of transportation policy - the ensuing crime effect - by examining its size, but also its nature, spatial concentration, speed of emergence, and its origins (i.e. new vs substituted). This detailed study will help with optimal policy design in the future. However, the more substantial contribution is to the economics of crime. Canonical models, such as Becker (1968), predict that criminals respond to opportunities such as those generated by the night tube. My results serve as a check of these predictions. However, there is very little research on the dynamics and particularly the frictions that govern the nature of the adjustment to the new equilibrium after a shock. I will show that judging by the evolution of the effect in space and time, these frictions are substantial and should not be ignored.

4 Data

I have data on all crimes recorded by all UK Police departments between January 2015 and December 2017. Each crime is recorded alongside its month of occurrence, its location, the type of crime, and a couple of other related factors. The locations are given in terms of coordinates, but to preserve identification of the crime, the exact coordinates can't be given. Instead, the UK Police keep a set of anonymous map points (e.g. half way down a street, the centre of a car park, etc), and they assign a crime the coordinates of the nearest map point. There are roughly 60,000 of these in my data across London; the implicit polygons which each point represents is therefore very small, creating no issues with analysis according to distance from a station. I will use these locations as my unit of observation.

I can also disaggregate the crime count into counts of different types of crime, which will be useful to investigate the effect in more detail. Certain mechanisms imply increases in certain types of crime, so by decomposing the effect across different crime types, I am able to better

understand the underlying mechanism behind the main effect.

Throughout my analysis, I work with the natural log of $1 +$ the crime count in a given location at a given month. The reason for this is that my data is heavily right-skewed, but there are many instances of zero crimes being committed in a location-month. Recent work by Chen and Roth (2024) has demonstrated that this transformation can harm analysis by introducing arbitrary unit-dependence, and I will address this issue in my analysis.

Importantly, the data gives all reported crimes, rather than just those which led to criminals being apprehended, which saves me from a large class of issues relating to measurement error and sample selection bias. The only possible issue of this nature is the potential increase in reporting due to the greater police presence at the stations, but I justify this isn't a huge issue above.

Alongside the crime data, I also have data on the location of each London Underground station. Geocoding these, and the crime locations above, allows me to work out the planar distance from each location to each tube station. After recording the lines that go through each station, I therefore know how far each crime location is from the nearest station served by a Night Tube, which is what I will use to define my baseline treatment.

Throughout my analysis, I keep only crime locations which are within 2km of a London Underground station. This is because urban and rural regions are fundamentally different and there is no reason to believe crime patterns evolve similarly in both. Doing this gives me access to a far more representative control group.

I also have data on all houses sold between 2015 and 2017 in England and Wales, including their sale price and address. I have geocoded these addresses, and determined the planar distance from each house sale location to each tube station less than 2km away as well, in the same way as above. I will use this for the house price analysis, if I decide to do it.

5 Methodology

In order to study the effect of the Night Tube on crime in surrounding regions, I must first define what it means for a crime location to be 'treated'. The task of defining treated units in situations where the treatment occurs at a point is a common problem in urban economics. In general, authors tend to define a unit as being treated if it is within a certain distance of the relevant point (e.g. Billings (2011), Zhang et al (2022)). I will follow this approach, using as my baseline definition of treatment that a location is 'treated' by the Night Tube if it is within 1km of an active Night Tube station. This distance threshold will be varied to ensure robustness, and I will also use alternative continuous treatment definitions to account for differing intensities.

To operationalise this, note that the staggered implementation of the Night Tube treatment allows me identify its effect on crime in surrounding areas using an event study. Let $y_{it} = \log(1 + c_{it})$, where c_{it} denotes the crime count at location i in month t . Define treatment as an indicator for whether the distance of location i at time t from the nearest active night-tube station is less than 1km: $D_{it} = \mathbb{1}\{\min_{m \in M_t} \text{dist}_{itm} \leq 1\}$, where dist_{itm} gives the planar distance from location i to underground station m , and M_t represents the set of such stations that have a Night Tube service running through them in month t . Then let s_i be the period of first

treatment, $s_i = \min\{t | D_{it} = 1\}$. The baseline event study specification I will use is then the following:

$$y_{it} = \alpha_i + \delta_t + \sum_{k \neq -1} \beta_k \mathbb{1}\{t - s_i = k\} + \Gamma \mathbf{X}_{it} + \epsilon_{it} \quad (1)$$

where in addition to the above, α_i and δ_t represent location and month fixed effects respectively, and $\mathbf{X}_{it}$ is a set of controls. The β_k coefficients represent the average treatment effect on the treated (ATT), under standard event study assumptions.

Beyond this, I will also decompose the treatment effect according to distance from the station, in order to understand its spatial decay. Formally, under the baseline definition of treatment at the 1km threshold, I divide the interval $[0, 1]$ up into N equally spaced intervals $\{b_1, \dots, b_N\}$, which represent distance bands - for example, $b_1 = [0, 0.25], \dots, b_4 = [0.75, 1]$. Then I can get distance-specific ATT evolutions by estimating the following adaptation of the specification above:

$$y_{it} = \alpha_i + \delta_t + \sum_{n=1}^N \sum_{k \neq -1} \beta_k^n \mathbb{1}\{t - s_i = k\} \mathbb{1}\{\min_{m \in M_t} \text{dist}_{itm} \in b_n\} + \lambda \mathbf{X}_{it} + \epsilon_{it} \quad (2)$$

The coefficient of interest here, β_k^n , represents the ATT at event-time k for locations in distance band b_n from the nearest active Night Tube station, again under standard assumptions.

To understand the underlying mechanisms behind the results observed from estimation of the specifications above, I will also use as alternative outcomes counts of specific types of crime. More formally, I will re-estimate the above specifications using as my dependent variable $y_{it}^j = \log(1 + c_{it}^j)$, where c_{it}^j gives the number of crimes of type j reported in location i and month t . UK Police data records 14 different crime types, including for example robbery, shoplifting, violence and sexual offences, and theft from the person.

I will also non-parametrically estimate the treatment effect, using a local linear regression. Specifically, I will regress y_{it} on the fixed effects and any controls, and then do a local linear regression on the residuals from this regression, which equation 1 suggests should roughly approximate the treatment effect.

I will also account for log(+1) issues coming from Chen and Roth (2024) by running some Poisson regressions and confirming I get similar results.

I plan to have the analysis run in the following order, with the following justification:

- Baseline treatment: within 1km
 - Baseline result: event study, with simple treatment and controls, to establish an effect
 - * TWFE OLS
 - * A+S
 - * BJS
 - These would all be reported together
 - Disaggregate by crime, to form an understanding of the mechanisms behind the observed effect
 - * Again all three estimators above, on the same graphs
 - Disaggregate by distance, to understand in detail the decay of the effect in space and whether crimes are being displaced or newly generated

- * Parametric, using the event study
 - TWFE OLS
 - BJS/A+S
- * Non-parametric, using kernel regression on errors
- Further investigation of the underlying motivation behind the spikes
 - * Investigate whether the crime spike around stations changes according to entry/exit rates at those stations - if it does this suggests the motivation is opportunity; if not then it is more likely to be mobility
 - * As above, but investigating whether it varies with the wealth of the area and particularly if this dependence emerges over time - if it does this suggests mobility is the motivation (facilitated by learning); if not then more likely to be opportunity
 - * Can also argue based on the individual crime patterns - house-based crimes should increase if the effect is mobility-driven, but not if opportunity-driven
- Poisson regression version of the above, to deal with Chen et al (2024) criticism of using $\log(+1)$
 - * TWFE (easy to do)
 - * BJS/A+S (harder - will need to code it up manually and maybe prove asymptotic normality of the coefficient estimates if I want to get SEs via bootstrapping)
 - * Maybe negative binomial too, to support
- Vary the sample, to demonstrate robustness: just locations within x km of a station for $x \neq 2$
- Vary the treated group, to investigate heterogeneity: just areas in inner/outer London
- Other forms of treatment
 - Vary the distance threshold
 - Continuous treatment - inverse distance
 - Continuous treatment - train/person arrival intensity
 - Continuous treatment - combination of the above

From these results, conclusions can be made about:

- The size of the crime effect
- The nature of the crimes that drive the effect
- The mechanism/motivation behind the effect
- The evolution of the effect in time
- The evolution of the effect across distance
- Whether the crime is new, or being displaced

6 Results

All of the following are under the baseline definition of treatment. I also have no controls yet other than my fixed effects - I will find and include these soon. First, results from estimating equation 1 via OLS and then using the estimator from Sun and Abraham (2021):

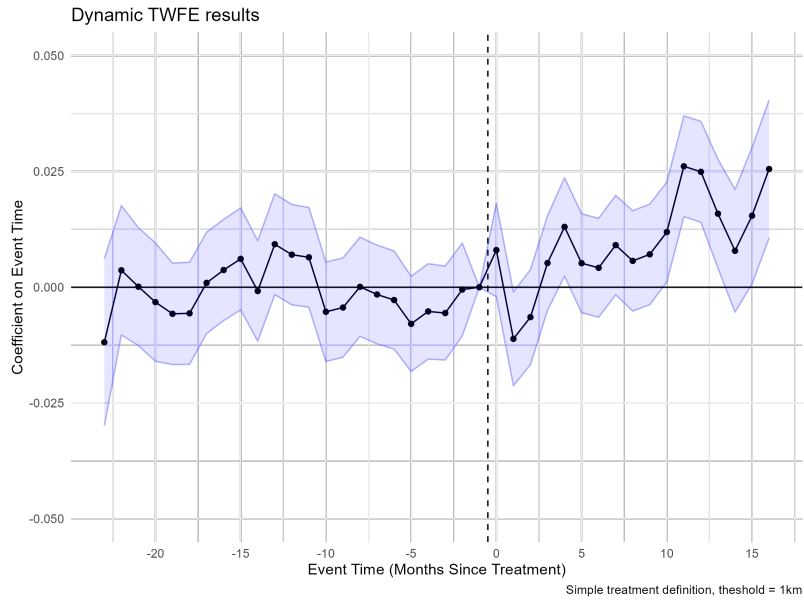


Figure 1: TWFE estimation of β_k from 1

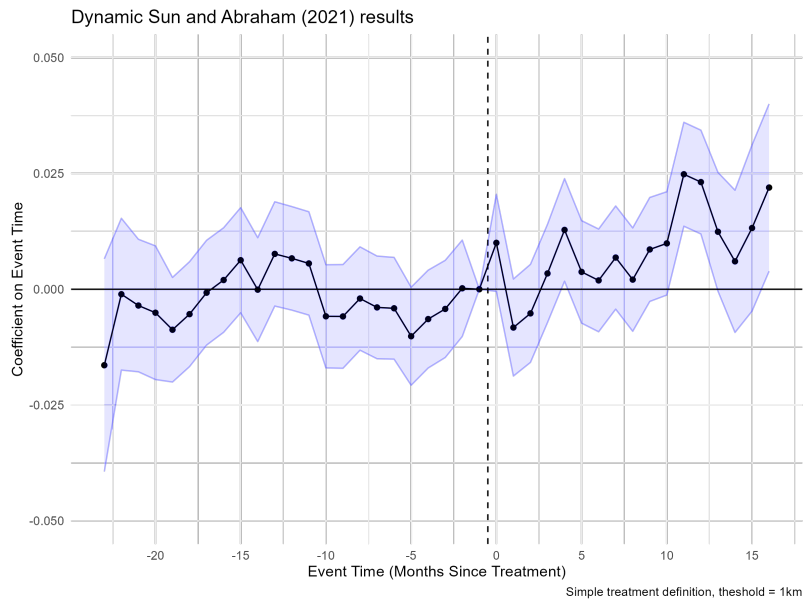


Figure 2: Sun and Abraham (2021) estimation of β_k from 1

Things to note:

- There appears to be a slight positive effect, that increases gradually over time
- There is no clear pre-trend, and the pre-treatment coefficients are generally insignificant, supporting the parallel trends assumption

Now investigate how each of the individual crimes evolves: estimate equation 1 for crime counts of each specific type of crime (i.e. the y_{it}^j):

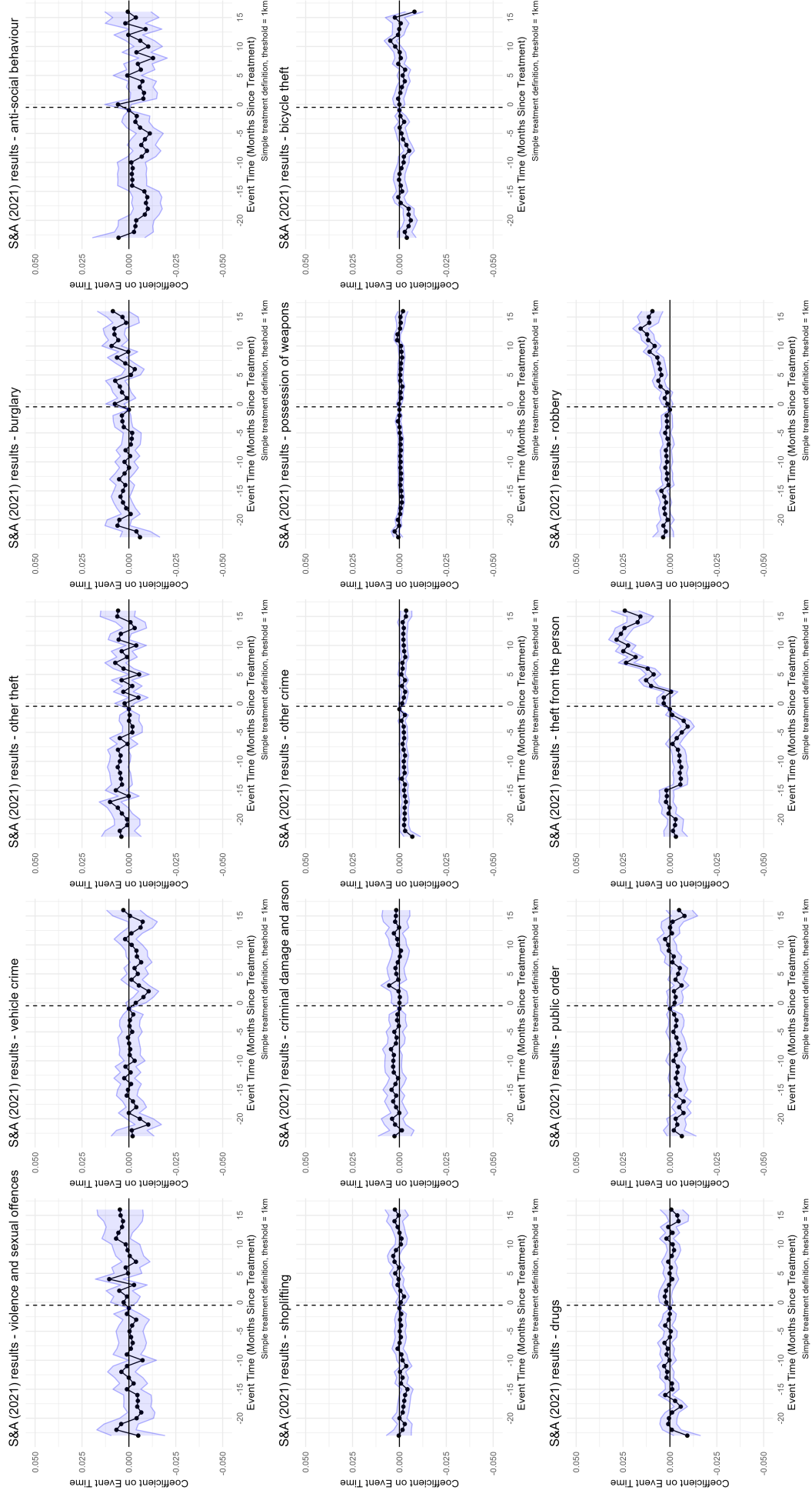


Figure 3: Sun & Abraham (2021) estimation of the β_k from equation 1 for each individual crime type

Things to note:

- All crime types exhibit a fairly clear null effect, apart from theft from the person, and robbery; both of which show clear gradual increases beyond the point of treatment
- The pre-treatment coefficients look good for every individual crime, which is reassuring
- Theft from the person is described as ‘crimes that involve theft directly from the victim but without the use or threat of physical force’. Robbery is described as ‘offences where a person uses force or threat of force to steal’. Robbery is therefore the more serious version of theft from the person, but both are when a criminal goes to a person on the street and takes some of their belongings. Implications:
 - We can now have an educated guess as to why crimes are increasing - more people are out walking at night, and are potentially easier targets due to being tired, alone and/or drunk, as well as a lack of natural light. Criminals are just making the most of this.
 - Also, seeing as robberies involve force and are therefore more serious, it is good for identification purposes that they increase too - this suggests the result is not just driven by an increase in the ease of reporting.
 - It is not surprising that theft from the person increases more than robbery. They are almost twice as common in the data, and it intuitively seems like it would be easier for a criminal to rob someone in this setting than usual, for the reasons above - they therefore have less need to resort to force.

With these results, and therefore a reasonable understanding of the size of the effect, its evolution over time, and the underlying mechanisms, I can now investigate how the effect evolves in space. To do this, I estimate the coefficients of equation 2 for the overall crime count, and for relevant crime-specific counts. I will use distance bands $b_1 = [0, 0.25], \dots, b_4 = [0.75, 1]$, splitting the treatment interval into 4 equally spaced subintervals. The results are as follows:

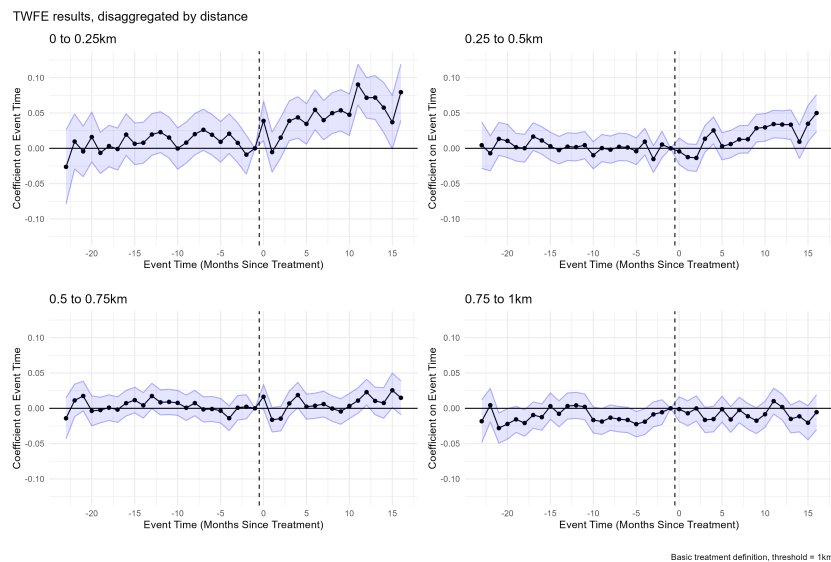


Figure 4: OLS estimation of β_k from equation 2, for the full crime count

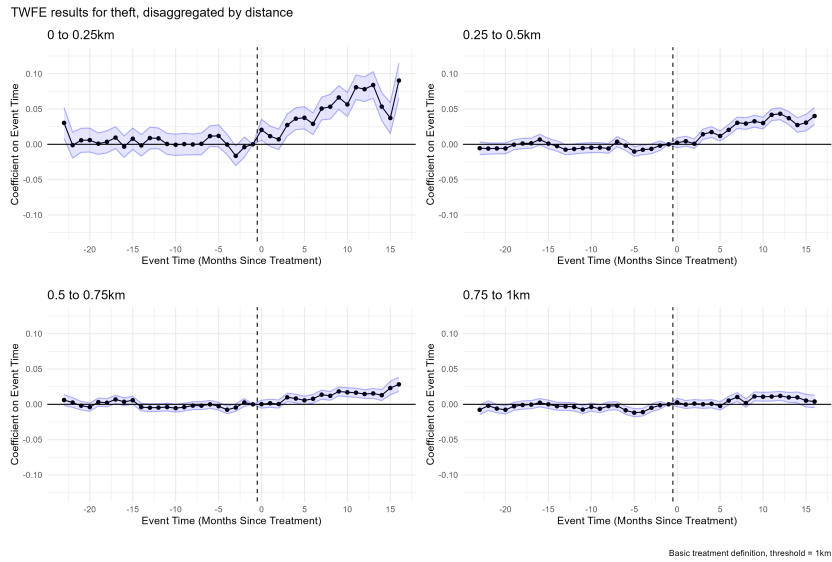


Figure 5: OLS estimation of β_k from equation 2, for thefts specifically

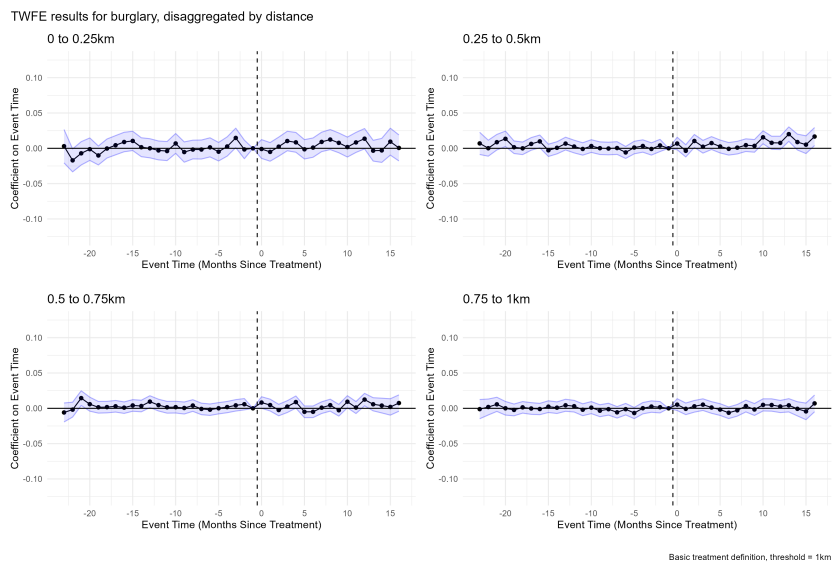


Figure 6: OLS estimation of β_k from equation 2, for burglaries specifically (as a placebo)

Things to note:

- I haven't got the one for robberies saved, but it has very similar patterns to thefts
- There is a clear spatial decay of the effect; by 0.75km-1km, there is almost no effect. This is expected - there is a higher concentration of 'targets' near the station, as from there people go in different directions
- In each distance band, we see the same time-evolution patterns as before, and the pre-treatment coefficients look good, which supports everything above.
- The 'irrelevant' crime types have a null effect at each distance band (burglary is the example shown)
- The presence of significant effects further out supports the justification that the reporting issue is not really a problem

I can also estimate the treatment effect non-parametrically, to get a better idea of the decay, as I described in the previous section. The plots of the estimated treatment effect *for theft from the person specifically*, and its evolution in time, are as follows:

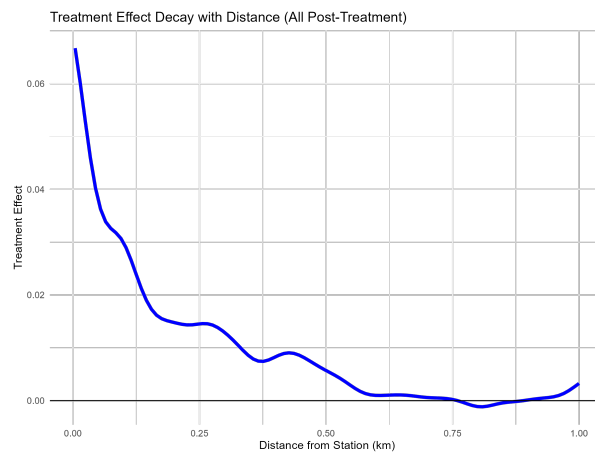


Figure 7: Non-parametric estimates of the ATT from local linear regression

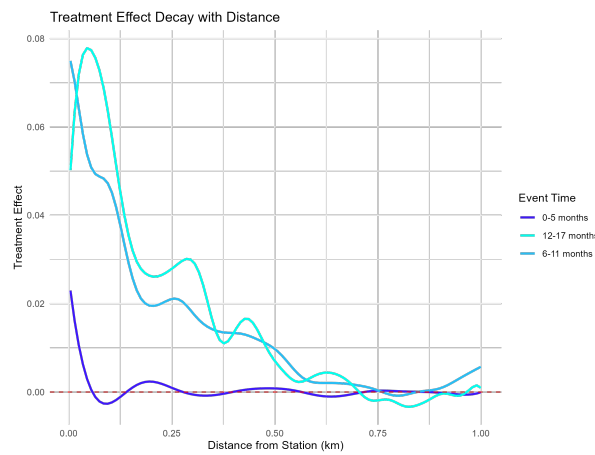


Figure 8: Evolution of the non-parametric ATT in time (6 month intervals)

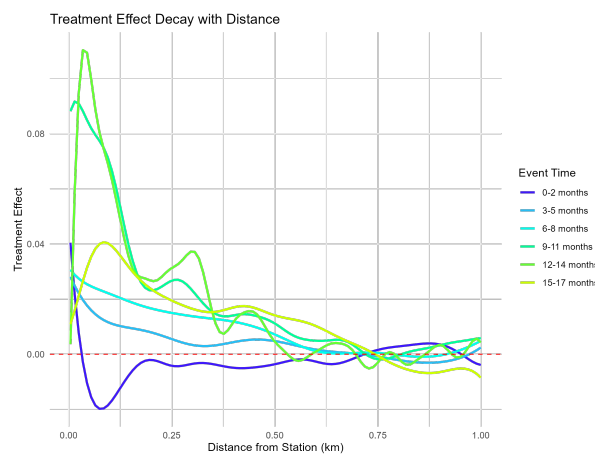


Figure 9: Evolution of the non-parametric ATT in time (3 month intervals)

Things to note:

- By 1km, the effect has disappeared
- There is nowhere where the treatment effect is negative - this suggests that the crimes are being newly generated (at least locally), rather than being substituted away from nearby areas
- The effect emerges gradually over time, but in general the profile over distance in each event-time period is similar

From the analysis above, my answers to the questions of interest above are:

- The crime effect is significant and positive
- It is driven by a surge in theft, likely due to the increased availability of opportunities from more people on the street susceptible to theft
- It isn't instantaneous - it grows over time, suggesting substantial criminal learning
- It decays over distance from the station, but is still significantly positive around 750m from the station
- It appears to be newly-generated crime, at least locally - there is no counteracting reduction in crime anywhere within 1km of the stations

7 Extensions

The things I still want to do are as follows:

- Do the Poisson regression to confirm the log(+1) issues aren't too big, and also attempt to prove the bootstrap is valid as a means of getting standard errors when using Poisson regression or non-parametric methods in BJS
- Do BJS estimation of everything above. I haven't done this yet because the R package keeps crashing. I think I'll just do it manually, but I need to write the code
- Do the robustness checks to more clearly establish that the fundamental driver of the effect is increased opportunity, rather than mobility or anything else
- Get the standard errors into the kernel regression - this was just taking a while in R so I haven't done it yet
- All the above with different definitions of treatment! I need to check it is robust to changes in distance and to the introduction of a continuous 'intensity' definition of treatment
- Maybe do the house price analysis
- Allow for spatial clustering of standard errors (e.g. Conley (1999))
- Maybe do triple differences for robustness: near vs far from the nearest station, treated vs untreated station, before vs after treatment
- Vary the sample (inner/outer London specifically, and varying my 2km cutoff)
- Get all my controls in

Beyond this, there are a few possible extensions I have in mind. I would like to work out whether there is crime displacement from other areas further than 1km from stations - I'm not sure how I'd do this, as parallel trends is unlikely to hold, but it would be very useful to know. I would also like to examine the welfare effects of the policy in more detail, to quantify the social cost of the crime effect, but I only have vague ideas about how I might implement this. As mentioned above, I am also very interested in working out whether Poisson regression or non-parametric regression can be used in BJS, from an econometric perspective.

One final thing I am interested in is as follows: I will address the mobility of criminals across different stations with my robustness checks above, but I am also interested in understanding the local mobility of criminals in these small areas surrounding stations. The question is as follows: are criminals living 1km away willing to move to the station in order to make the most of more crime opportunities there than at home? This is not a mechanism through which the Night Tube would lead to crime, but the exogenous variation in opportunities generated by the policy might allow me to analyse it.

Specifically, I can compare the observed decay in the treatment effect over distance to some benchmark decay under the assumption that criminals are perfectly immobile, if I know the distribution in space of both the origins of people entering/destinations of people exiting the stations at night, and of the locations of these fixed criminals. The challenge here is estimating these.

If I have these, I can then calculate the implied distribution of crimes, as follows (this could be wrong - I did it quickly, intuitively and late at night). First, assuming the region around a station is homogeneous conditional on distance from the station, the distributions only need to be specified on a line, rather than in a circle. For simplicity, consider distances on $[0, 1]$. Call the CDFs giving the distribution of criminals and of destinations across this interval F and G respectively, with PDFs f, g . Then for a given person travelling from the station to somewhere in $[0, 1]$, we have that very roughly speaking, the following holds:

$$\mathbb{P}(\text{person experiences crime at distance } d) = \mathbb{P}(\text{person reaches distance } d) \cdot \mathbb{P}(\text{crime at } d | \text{person reaches } d)$$

Under an assumption that the probability that a crime is committed is constant across criminals at different distances, this is just

$$\mathbb{P}(\text{person experiences crime at } d) \propto \mathbb{P}(\text{person reaches distance } d) \cdot \mathbb{P}(\text{criminal at } d)$$

Formally, if h is the density of crimes on $[0,1]$, then

$$h(d) = kG(d)f(d)$$

for some $k \in \mathbb{R}$

If for example we assume the distribution of criminals and of destinations is uniform in the unit circle around the station, then this implies once collapsing onto a line that $G(d) = 1 - d^2$, and that $f(d) = 1$. Thus $h(d) \propto 1 - d^2$. However, the density of criminals and particularly of destinations will be greater nearer 0, because the population density will be greater, and people are more likely to walk home if their home is closer to the station (noting that I am making an implicit assumption that all journeys are made on foot, due to the nature of the crime that increases). If I can estimate G and f , then I will get a curve which (after scaling) I can compare

with the output of the kernel estimation of the treatment effect, which will then allow me to comment on the extent of local criminal mobility.

I am currently in the process of working out how to estimate these densities. Criminal density should be fairly easy, but the distribution of destinations for people exiting the station on foot is challenging. I will think more though.

8 References

(for papers that might not be known)

Billings, Stephen B., Suzanne Leland, and David Swindell (2011). The Effects of the Announcement and Opening of Light Rail Transit Stations on Neighborhood Crime. *Journal of Urban Affairs* 33(5), 549-566.

Zhang, Y., Li, H. & Ren, G. (2022). Quantifying the social impacts of the London Night Tube with a double/debiased machine learning based difference-in-differences approach. *Transp. Res. Part A Policy Pract.* 163, 288–303.